Michael Mihaley

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Education

Columbia University New York, NY
Bachelor of Science in Computer Science, minor in Political Science *Sep. 2024 - May 2027*

City College of New York (Macaulay Honors College) New York, NY
Bachelor of Science in Computer Science, Dean's List *Sep. 2023 - May 2024 (transferred)*

Experience

Robotics and Machine Learning Researcher May 2025 – Sep. 2025
Columbia University, Advisors: Yunzhu Li and Yixuan Wang *New York, NY*

- Implemented Python/C++ control pipelines for 2 robotic systems (*Dexmate Vega, xArm7*), supporting joint-space and IK-based end-effector control with sub-cm positioning error
- Designed leader arm and extended the *GELLO Teleoperation Framework* to perform imitation reinforcement learning with the *Dexmate Vega*, simulating in *SAPIEN* and supporting 500 Hz control loops
- Designed and printed custom mounting connections to expand end effector options for the *Dexmate Vega* by 50%

Adjunct Faculty, Robotics Oct. 2025 – Present
Rudolf Steiner School *New York, NY*

- Teaching students advanced programming concepts such as finite state machines and multi-threading to push the boundaries of code execution time
- Instructing a *First Tech Challenge* robotics team across mechanical design, electronics, and Java programming, guiding them to achieve competition-ready results
- Scaled program size from 2-3 students to around 15 in one semester through hands-on curriculum design

Machine Learning Intern January 2024 – June 2024
Outlier AI *Remote*

- Refined large language models via reference-free evaluation in Python for a response accuracy increase of over 40%
- Built automated safety checks combining classifiers and rule tests to flag 10% of pre-deployed output as misaligned

Projects

Performance Optimized LLM Inference for Real-Time Robotic Planning | *Python, CUDA*

- Built a robotics planning pipeline using *Qwen2.5-3B-Instruct* to convert natural language commands into action sequences, conducted full-stack GPU profiling (compute vs memory-bound analysis).
- Utilized 16-bit quantization on weights and activations, VLLM prefix caching to reduced end-to-end latency by 60%, increased tokens/sec by 3×, and decreased VRAM usage by > 30%. Benchmarked run telemetry via *Weights and Biases.*
- Evaluated latency–throughput tradeoffs under single-query (robotics) vs batched (cloud) workloads, characterizing memory scaling with sequence length for embedded deployment feasibility.
- Built end-to-end pipeline integrating LLM inference with *ROS2* and *Gazebo* simulation for performing generated action sequences.

HybVIO Visual-Inertial Odometry System with SLAM | *C++, C*

- Extended a research-oriented odometry system that uses preprocessed data to interface with actual *Intel RealSense* Cameras for real-time spatial mapping under constrained compute (Quad-core, 1.2GHz clock speed)
- Collaborated with a team to identify slowdowns in stereo imaging code and speed up visualization by 60%

CUDA SHA-3 Hash Checker | *CUDA, C++, C*

- Developed an NVIDIA CUDA parallelized SHA-3 hasher and checker, allowing the SHA-3 hashing and comparison of over 14,000,000 keywords in under 10 seconds
- Wrote a memory-mapped file processor to reduce wordlist processing time from 15 seconds to 0.5 seconds when working with millions of lines of input data

Technical Skills

Languages: C++, Java, CUDA, Python, JavaScript, Kotlin, Typescript, C#
Development Tools: Docker, Android Studio, Fusion 360, SAPIEN, Drake, CMake, MuJoCo, Pytorch, Linux, ROS
Certifications and Fellowships: CAIAC AI Policy Fellow, Coursera Deep Learning Certified